

CORPORATE NETWORK & SYSADMIN COMMAND REFERENCE

CMD VS. POWERSHELL VS. LINUX BASH

Survival Guide for Senior Architects/SysAdmins - Troubleshooting

1. NETWORKS: BASIC CONNECTIVITY & DNS

PURPOSE (Purpose)	CMD WINDOWS (Legacy/Fast)	POWERSHELL (Modern/Object)	BASH LINUX (Universal)	NOTES (Veteran Notes)
Display Basic IP Information	ipconfig	Get-NetIPAddress - AddressFamily IPv4	ip addr show	Universal starting point. In modern Linux, forget <i>ifconfig</i> ; <i>iproute2</i> (<i>ip addr</i>) is the standard.
View Detailed Config (MAC/DNS)	ipconfig /all	Get-NetIPConfiguration -All	nmcli dev show	Essential to validate if DHCP delivered the correct Gateway and DNS suffixes for the VLAN.
Release / Renew DHCP	ipconfig /release ipconfig /renew	Restart-NetAdapter -Name "Eth0"	dhclient -r dhclient	Best Practice: Sometimes it's faster and cleaner to bounce the entire network interface (toggle) than wait for DHCP renegotiation.
Flush DNS Cache	ipconfig /flushdns	Clear-DnsClientCache	resolvectl flush-caches	Critical after server migrations or propagation failures. Requires elevation (Admin/Root).
DNS Lookup & Query	nslookup <host>	Resolve-DnsName <host>	dig <host>	The Linux <i>dig</i> command provides the most precise metadata (TTL, NameServers). In PS, <i>Resolve-DnsName</i> is excellent for scripting.

2. NETWORKS: ADVANCED TROUBLESHOOTING & ROUTING

PURPOSE (Purpose)	CMD WINDOWS (Legacy/Fast)	POWERSHELL (Modern/Object)	BASH LINUX (Universal)	NOTES (Veteran Notes)
Test Connection (Ping)	ping -t <host>	Test-Connection <host>	ping -c 4 <host>	Linux ping is infinite by default (use -c). Continuous ping (-t) in Windows saves lives when monitoring router reboots.
Trace Route (Path Discovery)	tracert <host>	Test-NetConnection - TraceRoute	mtr <host>	Senior Tip: <i>mtr</i> in Linux (My Traceroute) is unbeatable; it combines ping and traceroute in real-time on screen.
Check Open Ports & Sockets	netstat -ano	Get-NetTCPConnection	ss -tulpn	Identifies what is listening on the port. ss (Socket Statistics) replaced netstat in Linux because it is drastically faster.
Test Remote Port (Telnet-like)	<i>N/A (Requires Telnet Client)</i>	Test-NetConnection <IP> - Port <Port>	nc -zv <IP> <Port>	The Firewall trial by fire. Test-NetConnection (TNC) is the modern Windows Swiss Army knife. In Linux, use netcat (nc).
View Routing Table	route print	Get-NetRoute	ip route show	Mandatory diagnostic when the server has multiple NICs and traffic is leaving through the wrong gateway.

3. OPERATING SYSTEM: CPU & MEMORY

PURPOSE (Purpose)	CMD WINDOWS (Legacy/Fast)	POWERSHELL (Modern/Object)	BASH LINUX (Universal)	NOTES (Veteran Notes)
System & Kernel Info	systeminfo findstr "OS"	Get-ComputerInfo	uname -a / hostnamectl	First step when logging into an unknown server: know the exact build/kernel for compatibility.
Real-time CPU/RAM Usage	tasklist	Get-Process Sort CPU	top / htop	The Windows CLI is poor for real-time visualization. In Linux, <i>htop</i> is the SysAdmin's best friend for bottlenecks.
Check Free Memory	systeminfo find "Memory"	Get-CimInstance Win32_OperatingSystem	free -m / free -h	<i>free -h</i> (human readable) gives the answer in 1 second. Pay attention to the "available" column in Linux, not just "free".
System Uptime	systeminfo find "Boot Time"	(Get-CimInstance Win32_OperatingSystem).LastBootUpTime	uptime / w	Slow server? Check if it restarted on its own overnight (Crash/Updates) by validating the uptime.
Logged in Users	quser / qwinsta	Get-RDUserSession	who / w	Never restart a critical server without first checking if another analyst is logged into the console performing maintenance.

4. STORAGE, DISKS & PROCESSES

PURPOSE (Purpose)	CMD WINDOWS (Legacy/Fast)	POWERSHELL (Modern/Object)	BASH LINUX (Universal)	NOTES (Veteran Notes)
Disk Space Usage (Volumes)	wmic logicaldisk get size,freespace	Get-Volume / Get-PSDrive	df -h	The classic "Disk Full!". <i>df -h</i> in Linux and <i>Get-Volume</i> in PS are lifesavers in Sev 1 incidents.
Find Large Folders	dir /s /a /o-s	Get-ChildItem -Recurse Sort Length	du -sh * sort -h	Disk is full? Use <i>du -sh *</i> at the root to track down exactly which folder (usually logs) blew up.
Kill Hung Process	taskkill /F /PID <pid>	Stop-Process -Id <pid> - Force	kill -9 <pid>	The sledgehammer. The -9 signal (SIGKILL) in Linux and the /F flag in Windows kill the application mercilessly when it hangs the OS.
Check Service Status	sc query <service>	Get-Service <service>	systemctl status <service>	Modern Linux's <i>systemctl status</i> is brilliant because it displays the last error log lines of the service along with its status.
Restart Daemon/Service	net stop <svc> && net start <svc>	Restart-Service <service>	systemctl restart <service>	The most executed action in IT history. Restarting the service resolves 80% of application outages.